



COMBINED FOUNDATIONS

PURPOSE

This document establishes guidelines and recommended procedures for the design of combined foundation or mat supported on soil.

SCOPE

This document includes the following major sections:

- DEFINITIONS
- DESIGN CONDITIONS
- PEDESTAL DESIGN
- FOUNDATION SIZING
- SOIL BEARING
- FOUNDATION DESIGN
- REFERENCES
- ATTACHMENTS

APPLICATION

This document is to be used to design a single combined foundation or mat to carry all of the applied loads, whenever 2 or more pieces of equipment, building supports, or structure columns are so closely spaced that it is impractical or impossible to design individual foundations for each item.

DEFINITIONS

Combined Foundation: Any foundation supporting more than a single column or wall load.

Mat Foundation: A large concrete slab supporting a number of columns or pedestals.

DESIGN CONDITIONS

Applied Loads

The vertical and horizontal loads and overturning moments acting on the foundation will be the sum of all the vertical and horizontal loads and overturning moments from each individual item supported by the foundation. The loads from each item shall be determined in accordance with the procedure provided in the guideline governing the item being considered.



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Load Combinations

All the load combinations specified in governing codes, 000.215.STC01015 - Structural Design Criteria and any project specific requirements shall be considered in determining the design forces (shear forces and bending moments) as well as the soil bearing pressures for the foundation. Load reversal for seismic and wind loads should be considered in each of the principal areas of the foundation. In some cases, combining orthogonal seismic loads may be required. Refer to ASCE 7 – 10. Keep in mind, the critical combination of shear load and bending moment may not occur with the largest simultaneously applied load at each column or pedestal.

When hydrotest is a design consideration, the Design Engineer will verify whether the hydrotest will involve only 1 vessel at a time or if it is possible that more than 1 vessel can be simultaneously hydrotested. If more than 1 vessel supported on the same foundation is to be simultaneously hydrotested, the sequence of hydrotesting will be investigated and be taken into consideration.

Design Loads

Design loads (moments and shears) within the foundation will be determined by analysis using the appropriate load factors specified by ACI (American Concrete Institute) 318.

The design loads will not exceed the capacity of the concrete section as determined by the provisions of ACI 318.

The maximum applied soil bearing pressure will be computed using unfactored (working) loads and will not exceed the allowable soil bearing pressure recommended by the Geotechnical Engineer for that foundation.

PEDESTAL DESIGN

Pedestals and columns supported by the foundation, as well as the anchor bolts attaching the structures or equipment to the pedestals or columns, will be designed in accordance with the procedures provided in the practice for each particular item.

FOUNDATION SIZING

Center Of Gravity (CG)

The centroid of the contact area between the soil and the concrete foundation or mat will be set to coincide with the overall CG of all the sustained (long-term) loads, including the self-weight of the combined foundation or mat. This requirement, in combination with the arrangement of the various items being supported, will determine the minimum size and shape of the foundation or mat.



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In some instances, it may not be possible or practical to align the centroid of the foundation contact area with the CG of the sustained loads. In such a case, use engineering judgment and account for the effects of the resulting permanent eccentricity in the analysis and design of the foundation. Generally, due to the potential for differential settlements, a permanent eccentricity greater than 5 percent of the respective foundation dimension is considered to be unacceptable.

Thickness

The minimum foundation thickness will be as follows:

- 12 inches.
- Minimum thickness to develop the pedestal reinforcement.
- Minimum thickness required to resist 2-way (punching) shear without reinforcement at each pedestal or column.
- Minimum thickness to resist 1-way (diagonal tension) shear without reinforcement.
- Minimum thickness to ensure rigidity for manual analysis.

Note: *Whenever evaluating 2-way shear requirements, consider the effects of the applied moment at the base of the pedestal or column in addition to the effects of the applied vertical load; similar to the way it would be considered for slabs in ACI 318, Section 11.12.2.*

Depth

The depth of the foundation below finished grade is usually given in the project specifications based on recommendations from the Geotechnical Engineer. However, the Design Engineer will verify that the following conditions are met:

- Bottom of foundation is located below the frost level.
- Adequate clearance is provided between the top of the foundation and finished grade to allow placement of underground piping and electrical.
- Sufficient embedment is provided to ensure overall stability.

Stability Ratio

The minimum stability ratio (minimum resisting moment/overturning moment) for all unfactored load combinations will be as specified in the project specifications.

SOIL BEARING

Except for extremely unusual problems, assume the soil bearing pressure at the base of a combined foundation to follow either a distribution governed by elastic subgrade reaction (nonlinear) or a straight line (linear) distribution. In either case, the resultant of the soil bearing pressure will coincide in location, and will be equal and of opposite direction to



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the resultant of the applied loads. At no place will the calculated soil bearing pressure exceed the maximum allowable soil bearing pressure.

For large combined foundations and mats, the allowable soil bearing pressure will be determined with consideration made for potential differential settlements with respect to adjacent structures or foundations.

The allowable soil bearing pressure will not be increased for any load combination unless specifically allowed by the project specification.

Linear Distribution

Assume a linear soil bearing pressure distribution for all foundations that can be considered rigid; that is to say, very small deformations will result from the applied loading. Criteria as to whether or not a foundation is rigid are given below.

Assume a linear soil bearing pressure distribution in cases where the footing design is governed by shear stresses.

Whenever an item, such as a tower, contributes virtually all the load and moment affecting the foundation, the foundation is normally considered rigid, and a linear distribution of soil pressure is assumed.

If two or more items such as two vertical vessels contribute approximately equal loads and moments to the foundation, the engineer may assume a rigid foundation for the purpose of determining maximum allowable soil bearing pressure. However, use a strip analysis or similar procedure in determining the footing stresses.

Nonlinear Distribution

A nonlinear soil bearing pressure distribution, found by means of subgrade reaction theories, is based on a simplification of the complex relationship between the bearing pressures and the corresponding soil deformations. If applied to subsoils of such character that the deformations are localized in the general vicinity of the loads, and if the maximum bearing pressures are smaller than about 1/2 the ultimate bearing capacity of the soil, this analysis provides reasonably accurate results.

The modulus of subgrade reaction depends on many variables and should be provided by the Geotechnical Engineer in the site specific soil report. Refer to Attachment 01, Table 1, for a range of typical values for use in preliminary designs and estimates.

Refer to Bowles' "Foundation Analysis and Design," for a general discussion on the modulus of subgrade reaction and the effects of foundation shape, size, and depth.



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FOUNDATION DESIGN

Rigidity Check

The following criteria to determine if a foundation is rigid are based on ACI 336.2R, "Suggested Analysis and Design Procedures for Combined Footings and Mats," and the "Foundation Engineering Handbook."

A foundation will be considered rigid if the relative stiffness factor, K_r , is greater than or equal to 0.5.

The relative stiffness factor, K_r , compares the overall stiffness of the foundation, structural framing members, and shear walls with that of the underlying soil.

To determine the relative stiffness factor, make an analysis comparing the combined stiffness of the foundation, superstructure framing members, and shear walls with the stiffness of the soil. That is,

$$K_r = \frac{E' I_B}{E_s B^3} \quad \text{Equation 1 (ACI 336.2R, Eq 5-1)}$$

$$E' I_B = \left[E' I_F + \sum E' I_B + \sum \frac{E' t_w h_w^3}{12} \right] \quad \text{Equation 2 (ACI 336.2R, Eq 5-2)}$$

Where:

E' = Modulus of elasticity of the materials used in the respective portions of the structure and foundation (ksf)

I_b = Moment of inertia of the structural framing members (ft^4)

I_F = Moment of inertia of the foundation (ft^4)

E'_s = Modulus of elasticity of the soil (ksf)

a = Thickness of shear wall (ft)

h = Height of shear wall (ft)

b = Width of footing (ft)



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It would be conservative to ignore the stiffness of the structural framing members and shear walls when computing K_r .

The modulus of elasticity of the soil, E'_s , should be provided by the Geotechnical Engineer in the site specific soil report. Refer to Attachment 01, Table 2, for a typical range of values for use in preliminary designs and estimates.

Alternatively, a foundation or mat will be considered rigid if the pedestal or column loads and spacings are relatively uniform (vary by no more than 20 percent) and the pedestal or column spacing is less than,

$$1.75 \sqrt{\left(\frac{k_b b}{4E_c I} \right)^{1/4}} \quad \text{Equation 3 (ACI 336.2R, Eq 5-3)}$$

Where

k_b = Modulus of subgrade reaction (kcf)

b = Width of a strip of foundation or mat between centers of adjacent bays (ft)

E_c = Modulus of elasticity of concrete (ksi)

I = Moment of inertia of a foundation strip of width b (ft⁴)

Analysis and Design

Rigid Foundations

Analyze foundations and mats that have been determined to be rigid on the basis of simple static and rigid body mechanics.

Once the soil bearing has been determined, develop a shear and moment diagram for each principal direction of the foundation.

Take the critical sections for moment and shear with respect to the face of a square of area equivalent to that of the major pedestal. If there is not a major pedestal, take the critical sections for moment and shear at the centerline of the pedestal.

Determine the external moment on any section by passing through the section a vertical plane that extends completely across the footing, and compute the moment of the forces acting over the entire area of the footing on one side of the plane.

Check shear as a measure of diagonal tension according to ACI 318.



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Place the bottom reinforcing required at the critical sections continuously across the entire footing. If possible, avoid variations in reinforcing sizes for opposite directions to prevent errors in placement during construction.

Top steel, as required for concentrated loads and uplift, may or may not be distributed across the entire footing width depending on each individual case.

Strip Analysis

Analyze rigid foundations with more than one major load source using a strip analysis as outlined below.

Strips should be taken across the entire footing in the area of the major loads. The strip width will normally be the width of the equivalent square pedestal and 3 times the footing thickness, or 1-1/2 the footing thickness on each side of the equivalent square.

Next, determine the soil bearing pressures required to produce equilibrium with all the loads and moments acting on the strip.

Determine critical shear and moment on the basis of simple statics, and place the resulting reinforcing continuously across the entire footing.

Flexible Foundations

The analysis and design of a nonrigid (flexible) foundation or mat is very rigorous and time consuming. A description of various methods such as beam or elastic foundation can be found in soils and foundation textbooks. Because of the difficulty and time involved, computerized methods should be used for the analysis of flexible foundations or mats.

Once the foundation has been accurately modeled and analyzed, the location and magnitude of the critical moments and shears can be determined. From this, the required foundation thickness and reinforcement can be evaluated.

Computer Analysis

Rigid Foundations

A 2-dimensional structural analysis program can be used to analyze a rigid foundation or mat.

The applied loading and resulting soil bearing pressure will be determined for each load combination to be considered. The foundation will be modeled with beam elements and all nodes except 1 are free to rotate and translate as controlled by the stiffness of the beam elements.



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Overall stability is provided by fixing a node at one end of the foundation against any translations and rotations. Resulting forces and moments at this node provide a measure of the imbalance between the applied loads and the calculated soil bearing pressure used in the analysis.

The magnitude of these unbalanced forces and moments should be small in comparison to the moments and shears used for the design of the foundation.

The moments and shears determined by the analysis are used to develop the thickness and reinforcing requirements discussed above.

Flexible Foundations

Before starting a computerized analysis of a flexible foundation or mat, the Design Engineer will investigate which computer programs are available.

There are a great number of variables and uncertainties that must be addressed when creating a model for a computerized analysis of a flexible foundation or mat. Therefore, attention to details and good engineering judgment must be exercised.

Use conservative assumptions and work closely with the Geotechnical Engineer to develop realistic values for the stiffness of the soil and do not rely on values taken from textbooks.

The Design Engineer will have thorough understanding of the model and how the software will interpret the model so that a realistic solution can be obtained.

RESOURCES

Building Codes and Other Design Resources

PCA Notes on ACI 318 (Current Version). Building Code Requirements for Structural Concrete

ACI Concrete Design Handbook SP-17, 1997

IBC (Current Version) - International Building Code. International Code Council

Hsai-Yang Fang. Foundation Engineering Handbook. Van Nostrand Reinhold, New York, NY, Second Edition, 1991



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REFERENCES

Civil Structural Engineering:

[000.215.STC01015](#) Structural Design Criteria

Building Codes and Other Design Resources

ACI 336.2R-02	Suggested Analysis and Design Procedures for Combined Footings and Mats. American Concrete Institute. Farmington Hills, Michigan. 1993
ACI 318	Building Code Requirements for Structural Concrete and Commentary. American Concrete Institute. (Current Version)
Joseph E. Bowles.	Foundation Analysis and Design. McGraw-Hill Inc. New York, NY. Fifth Edition, 1996
ASCE 7	Minimum Design Loads for Buildings and Other Structures. American Society of Civil Engineers, (Current Version)

ATTACHMENTS

Attachment 01	Range Of Typical Soil Values
Attachment 02	Sample Design Of A Combined Foundation
Attachment 03	Fig 16-13 from PCA Notes on ACI 318-05 (This is referred in attachment 02 calculations)

COMBINED FOUNDATIONS**Range of Typical Soil Values****Table 1: Range of Values of Modulus Of Subgrade Reaction (K_s), (reprinted from Bowles)**

Soil	Range of K_s , kcf
Loose sand	30 - 100
Medium sand	60 - 500
Dense sand	400 - 800
Clayey sand (medium)	200 - 500
Silty sand (medium)	150 - 300
Clayey soil (q_u in ksf):	
$q_u < 4$	75 - 150
$4 < q_u < 8$	150 - 300
$8 < q_u$	> 300

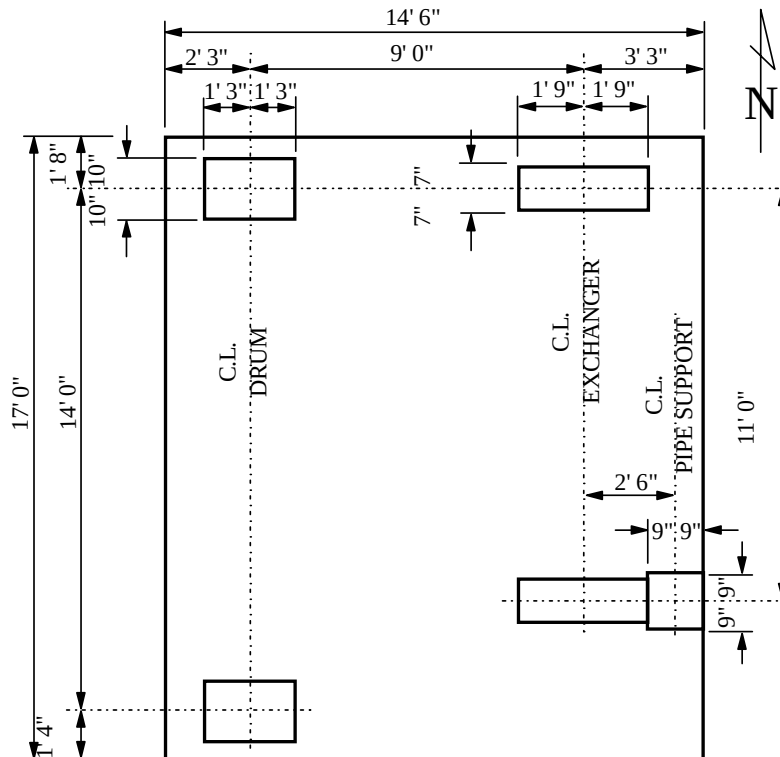
These values should be used for a guide. Project specific values may be higher or lower and should be obtained from Geotechnical Engineer.

Table 2: Typical Range Of Values For Modulus Of Elasticity Of Soil E'_s , (reprinted from Fang)

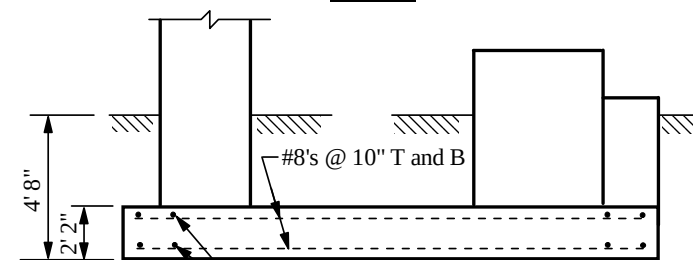
Type of Soil	Modulus of Elasticity, psi
Very soft clay	50 - 400
Soft clay	250 - 600
Medium clay	600 - 1,200
Hard clay	1,000 - 2,500
Sandy clay	4,000 - 6,000
Silty sand	1,000 - 3,000
Loose sand	75 - 150
Dense sand	150 - 300
Dense sand and gravel	> 300

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Sample Design of A Combined Foundation



PLAN



SECTION

#8's @ 10" T and B

Note: Drum and exchanger pier dimensions are the same at each end. Details and reinforcing are not shown. Slide plate details are not shown.



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Sample Design of A Combined Foundation

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Sample Design of A Combined Foundation

Design Data

Concrete:

28 Day Compression Strength, $f'_c = 4,000$ psi
ASTM 615 Grade 60 deformed bars, $f_y = 60,000$ psi
Reinforced concrete capacity determined per the provisions of ACI 318-05

Soil: (medium clay)

Allowable soil bearing = 3,500 psf net (soil report does not allow a 1/3 increase for seismic or wind)
Modulus of Elasticity, $E_s = 1,000$ psi
Modulus of Subgrade Reaction, $K_s = 100$ kcf
Soil Density, $\gamma = 100$ pcf

To allow placement of underground piping and electrical provide 2'-6" of earth cover over foundation.

The water table is well below the bottom of foundation.

Sample Design of A Combined Foundation

The diagram shows an elevation view of a foundation mat and retaining wall. The foundation mat is a horizontal rectangular base. Above it, on the left, is a vertical wall with a T-shaped top. The total height of the wall from the top of the foundation mat to the top of the T-shape is 16' 0". A break symbol (two parallel diagonal lines) is shown on the wall. To the right of the wall, the ground surface is indicated by a horizontal line with diagonal hatching below it, labeled "Grade". The vertical distance from the top of the foundation mat to the "Grade" line is 2' 6". To the right of the "Grade" line, there is a rectangular structure with a height of 5' 0". Further right, there is a smaller rectangular structure with a height of 3' 0". A downward arrow labeled "+P" is shown to the right of the 3' 0" structure. A red horizontal line at the bottom of the foundation mat is labeled "Top of Foundation Mat".



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Sample Design of A Combined Foundation

Summary of Loads Applied at top of Foundation Mat

		Empty			Operating			
		Eqpt & Pedestal	Seismic* N to S	Seismic* W to E	Eqpt & Pedestal	Friction (growth)	seismic* N to S	seismic* W to E
Drum (anchor end)	P	38 kips	-7.2 kips	-	60 kips	-	-13.7 kips	-
	V	-	22.9 kips	14.0 kips	-	-4.6 kips	39.1 kips	22.1 kips
	M	-	338.7 ft-k	246.3 ft-k		-73.6 ft-k	597.9 ft-k	421.7 ft-k
Drum (sliding end)	P	38 kips	7.2 kips	-	60 kips	-	13.7 kips	-
	V	-	5.2 kips	14.0 kips	-	4.6 kips	5.2 kips	22.1 kips
	M	-	55.5 ft-k	246.3 ft-k	-	73.6 ft-k	55.5 ft-k	421.7 ft-k
Exchanger (anchor end)	P	35.2 kips	-13.9 kips	-	46 kips	-	-18.9 kips	-
	V	-	30.2 kips	7.0 kips	-	-6.4 kips	40.5 kips	9.2 kips
	M	-	147.3 ft-k	66.0 ft-k	-	-32 ft-k	198.8 ft-k	89.8 ft-k
Exchanger (sliding end)	P	50.8 kips	13.9 kips	-	68 kips	-	18.9 kips	-
	V	-	0.8 kips	10.1 kips	-	6.4 kips	0.8 kips	13.5 kips
	M	-	2.0 ft-k	98.1 ft-k	-	32 ft-k	2.0 ft-k	133.2 ft-k
Pipe Support	P	2 kips	0	0	3 kips	-	0	0
	V	-	0.2 kips	0.2 kips	-	-	0.3 kips	0.3 kips
	M	1 ft-k	3 ft-k	3 ft-k	1.5 ft-k (E/W)	-	4.5 ft-k	4.5 ft-k

Notes:

1. Lateral and friction loads shown reflect possible load reversals. Positive load directions are defined on page 4.
2. Loads include weight of column or pier.

The details of vessel pier, exchanger pier and seismic calculation are omitted for brevity. For information on this design, refer to practices PIP guideline STE 03350, STE 03360, and FD guideline 000.215.1216 respectively.

In this example seismic loads control. If wind loading is a significant consideration, then latter must be included.

*** Earthquake forces are calculated based upon strength level consideration**



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Sample Design of A Combined Foundation

Sizing of Foundation/ Unfactored Loads

For center of gravity calculations, use centerline of drum at anchor end as origin as indicated on layout.

Determine Center of Gravity (Operating)

	P	Soil Displaced by Pier	P _{net}	North - South Direction		East - West Direction	
				Y	P _{net} Y	X	P _{net} X
Drum (anchor end)	60.0 kips	1 kip	59.0 kips	0	0	0	0
Drum (sliding end)	60.0 kips	1 kip	59.0 kips	14 ft	826 ft-k	0	0
Exchanger (anchor end)	46.4 kips	1 kip	45.4 kips	0	0	9 ft	409 ft-k
Exchanger (sliding end)	67.6 kips	1 kip	66.6 kips	11 ft	733 ft-k	9 ft	599 ft-k
Pipe Support	3.0 kips	0.6 kip	2.4 kips	11 ft	26 ft-k	11.5 ft	28 ft-k
Totals =			232.4 kips		1,585 ft-k		1,036 ft-k

$$Y_{na} = \sum (P_{net} Y) / \sum (P_{net}) = (1,585 \text{ ft-k}) / (232.4 \text{ kips}) = 6.82 \text{ ft}$$

$$X_{na} = \sum (P_{net} X) / \sum (P_{net}) = (1,036 \text{ ft-k}) / (232.4 \text{ kips}) = 4.46 \text{ ft}$$

Determine Center of Gravity (Empty)

Calculations are omitted for brevity. The results are:

$$P_{net} = 159.4 \text{ kips} \quad Y_{na} = 6.78 \text{ ft}, \quad X_{na} = 4.84 \text{ ft}$$

Trial Mat Size and Location Based on the Operating C. G. (refer to sketch on next page)

Note!!! This sample calculation designs the mat as a rigid body in order to simplify the calculation. A thinner mat could be designed using a flexible mat analysis. A flexible mat analysis would normally require a finite element computer analysis.

Determine Mat Thickness Required for Rigid Behavior

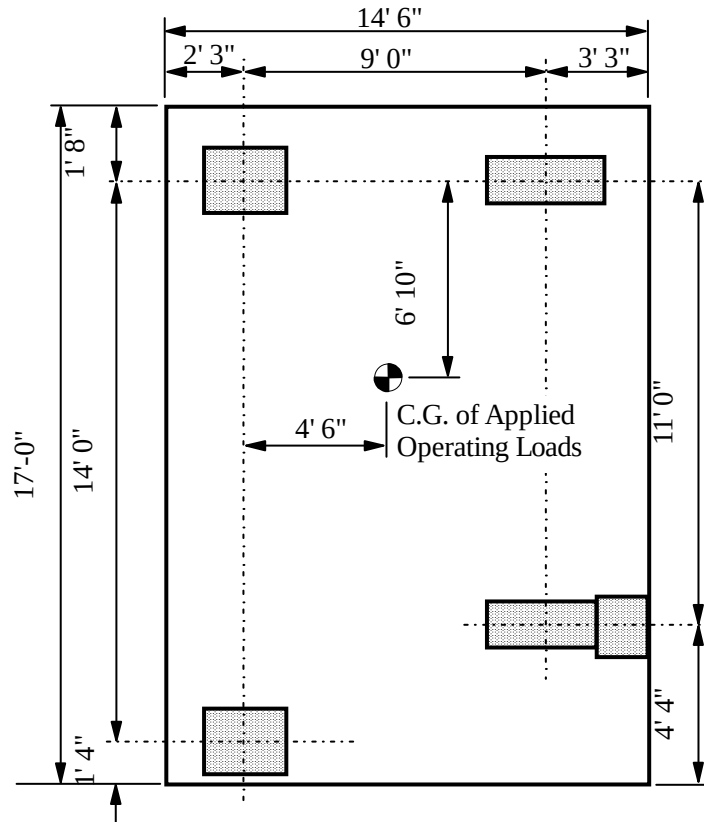
Concrete Modulus of Elasticity, (ACI 318, section 8.5.1)

$$E_c = 57,000 \sqrt{f'_c} = 57,000 \sqrt{4000 \text{ psi}} = 3,605,000 \text{ psi, or } 3,605 \text{ ksi}$$

Soil modulus of elasticity (from design data), $E'_s = 1,000 \text{ psi, or } 1 \text{ ksi}$

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Sample Design of A Combined Foundation



Length of mat, $b = 17.0$ ft

Unit moment of inertia of foundation ignoring stiffness of vessels (from Practice), $I_B = (b_f t^3 / 12) / b_f$

Stiffness criteria (from Practice), $K_r = E_B I_B / E_S b^3$

Rearranging stiffness equation,

$$I_B > (0.5 E_S b^3) / E_B$$



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Sample Design of A Combined Foundation

Substituting for I_B

$$t_{req} = [6 E'_s b^3 / E_B]^{1/3} = [6 (1 \text{ ksi})(17 \text{ ft})^3 / (3,605 \text{ ksi})]^{1/3} = 2.01 \text{ ft} > 12 \text{ in minimum}$$

Pier reinforcing development should also be considered. USE 2'-2" THICKNESS

Check Offset Criteria

By inspection of trial mat location, the east/west direction is not centered

	density times dimensions	P_{net}	East - West Direction	
			X	$P_{net} X$
Operating Load Results	-	232.4 kips	4.46 ft	1,036 ft-k
Foundation Mat	(0.15 kcf) (14.5')(17')(2.17')	80.2 kips	5 ft	401 ft-k
Soil over Mat	(0.1 kcf)(14.5')(17')(2.5')	61.6 kips	5 ft	308 ft-k
Totals =	-	374.2 kips	-	1,745 ft-k

$$X_{na} = \sum (P_{net} X) / \sum (P_{net}) = (1,745 \text{ ft-k}) / (374.2 \text{ kips}) = 4.66 \text{ ft}$$

$$\text{offset} = (\text{distance to mat center}) - (X_{na}) = (5 \text{ ft}) - (4.66 \text{ ft}) = 0.34 \text{ ft}$$

$$\text{percentage offset} = 100 [(0.34 \text{ ft}) / (14.5 \text{ ft})] = 2.3 \% < 5 \%, \text{ OK}$$

Check Overturning Stability

By inspection of the applied loads, Empty plus East to West Seismic controls

$$U = 0.9D + 0.7E \quad (\text{Refer Structural Design Criteria, Table 5 – PIP STC 01015})$$

Overturning Moment, (in this case, all overturning moments are seismic)

$$\begin{aligned} \text{OTM} &= 0.7 * (\Sigma M + \Sigma V t + \Sigma P d) \quad (\text{note that vertical loads cancel out}) \\ &= 0.7 * [\{2 (246.3 \text{ ft-k}) + (66 \text{ ft-k}) + (98.1 \text{ ft-k}) + (3 \text{ ft-k})\} + \{2 (14.0 \text{ k}) + (7 \text{ k}) + (10.1 \text{ k}) + (0.2 \text{ k})\} (2.17 \text{ ft})] \\ &= 530 \text{ ft-k} \end{aligned}$$

Resisting Moment, (in this case, all resisting moments are dead weight)

$$\begin{aligned} \text{RM} &= (\text{mat and soil weight})(\text{mat width}) / 2 + (\text{applied weight})(\text{distance to empty load C.G.}) \\ &= (80.2 \text{ kips} + 61.6 \text{ kips})(14.5 \text{ ft}) / 2 + (159.4 \text{ kips})(2.25 \text{ ft} + 4.84 \text{ ft}) \\ &= 2,158 \text{ ft-k} \end{aligned}$$



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Sample Design of A Combined Foundation

Stability Ratio,

$$SR = 0.9 \text{ RM} / \text{OTM} = (0.9)(2,158 \text{ ft-k}) / (530 \text{ ft-k}) = 3.66 > 1.0 \text{ OK}$$

Note that a condition of one empty vessel plus one full vessel must be investigated if applicable.

Check Sliding Stability

Not required for seismic load combinations. Refer to Guideline 000.215.1216.

Check Soil Bearing

$$\text{area of mat, } A = (14.5 \text{ ft})(17 \text{ ft}) = 247 \text{ sf}$$

By inspection of the applied loads, Operating plus East to West Seismic controls

$$U = D + 0.7E \quad (\text{Refer Structural Design Criteria, Table 5 – 000.215.STC01015})$$

$$P = (\text{mat and soil weight}) + (\text{operating weight}) = (80.2 \text{ kips} + 61.6 \text{ kips}) + (232.4 \text{ kips}) = 374.2 \text{ kips}$$

$$V = 2 (22.1 \text{ kips}) + (9.2 \text{ kips}) + (13.5 \text{ kips}) + 0.3 \text{ kips} = 67.2 \text{ kips}$$

$$\begin{aligned} M &= 0.7 * [\Sigma (\text{seismic moments}) + (V)(\text{mat thickness})] + (\text{operating weight})(\text{C.G. offset}) \\ &= 0.7 * [2 (421.7 \text{ ft-k}) + (89.8 \text{ ft-k}) + (133.2 \text{ ft-k}) + (4.5 \text{ ft-k}) + (67.2 \text{ kips})(2.17 \text{ ft})] + [(232.4 \text{ kips})(0.34 \text{ ft})] \\ &= 931 \text{ ft-k} \end{aligned}$$

eccentricity,

$$e = M / P = (931 \text{ ft-k}) / (374.2 \text{ kips}) = 2.49 \text{ ft} > b/6$$

maximum gross soil bearing,

$$SB_{\max} = 2 P / 3 a [(b/2) - e] = 2 (374.2 \text{ kips}) / 3 (17.0 \text{ ft}) [(14.5 \text{ ft}) / 2 - (2.49 \text{ ft})] = 3.08 \text{ ksf}$$

allowable gross soil bearing,

$$SB_{\text{allow}} = (SB_{\text{net}}) + (\text{soil density})(\text{depth}) = (3.5 \text{ ksf}) + (0.1 \text{ kcf})(2.5 \text{ ft} + 2.17 \text{ ft}) = 4.0 \text{ ksf} > SB_{\max}, \text{ OK}$$

USE 17'-0" BY 14'-6" BY 2'-2" MAT

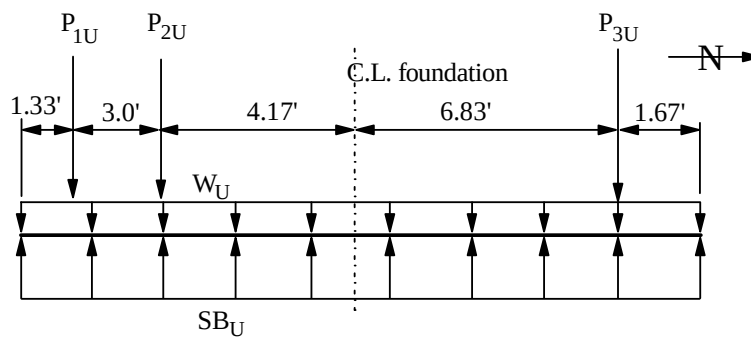
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Sample Design of A Combined Foundation

COMPUTATION OF FORCES FROM FACTORED LOADS

Load cases in the north/south direction are illustrated in this sample calculation. East/west loads, and associated foundation design, must be included in an actual design.

Case #1, Operating



$$U = 1.4 D$$

(Refer Structural Design Criteria, Table 6 – PIP STC 01015)

mat & soil weight,

$$w_u = 1.4 (14.5 \text{ ft width}) [(0.15 \text{ kcf concrete})(2.17 \text{ ft}) + (0.10 \text{ kcf soil})(2.5 \text{ ft})] = 11.68 \text{ klf}$$

sliding end of drum,

$$P_{1U} = 1.4 (P_{\text{net}}) = 1.4 (60 \text{ kips}) = 84 \text{ kips}$$

sliding end of exchanger & pipe support,

$$P_{2U} = 1.4 [\sum P_{\text{net}}] = 1.4 [(68 \text{ kips}) + (3 \text{ kips})] = 99.4 \text{ kips}$$

anchored end of drum & exchanger,

$$P_{3U} = 1.4 [\sum P_{\text{net}}] = 1.4 [(60 \text{ kips}) + (46 \text{ kips})] = 148.4 \text{ kips}$$

$$P = \sum P_u = (11.68 \text{ klf})(17 \text{ ft}) + (84 \text{ kips}) + (99.4 \text{ kips}) + (148.4 \text{ kips}) = 530.4 \text{ kips}$$

summed loads on foundation,



COMBINED FOUNDATIONS

Sample Design of A Combined Foundation

$$M = \sum (P_u)(d) = (84 \text{ kips})(7.17 \text{ ft}) + (99.4 \text{ kips})(4.17 \text{ ft}) + (148.4 \text{ kips})(-6.83 \text{ ft}) = 3.2 \text{ ft-k}$$

eccentricity,

$$e = M / P = (3.2 \text{ ft-k}) / (530.4 \text{ kips}) = 0.006 \text{ ft} \text{ too small to be of significance, neglect}$$

maximum soil bearing,

$$SB_{\max} = P / L = (530.4 \text{ kips}) / (17 \text{ ft}) = 31.2 \text{ klf}$$

at point #1:

$$V_{UL} = (31.2 \text{ klf} - 11.68 \text{ klf})(1.33 \text{ ft}) = 26.0 \text{ kips}$$

$$V_{UR} = (26.0 \text{ kips}) - (84 \text{ kips}) = -58.0 \text{ kips}$$

$$M_u = (26.0 \text{ kips})(1.33 \text{ ft}) / 2 = 17.3 \text{ ft-k}$$

at point #2:

$$V_{UL} = (-58.0 \text{ kips}) + (31.2 \text{ klf} - 11.68 \text{ klf})(3 \text{ ft}) = -58.0 \text{ k} + 58.6 \text{ k} = 0.6 \text{ kips}$$

$$V_{UR} = (0.6 \text{ kips}) - (99.4 \text{ kips}) = -98.8 \text{ kips}$$

$$M_u = (31.2 \text{ klf} - 11.68 \text{ klf})(4.33 \text{ ft})^2 / 2 - (84 \text{ kips})(3 \text{ ft}) = -69.0 \text{ ft-k}$$

between point #2 and #3:

$$V_u = 0 \text{ at } X = (98.8 \text{ kips}) / (31.2 \text{ klf} - 11.68 \text{ klf}) = 5.06 \text{ ft}$$

$$\begin{aligned} M_{\max} &= (31.2 \text{ klf} - 11.68 \text{ klf})(4.33 \text{ ft} + 5.06 \text{ ft})^2 / 2 - (84.0 \text{ kips})(3 \text{ ft} + 5.06 \text{ ft}) - (99.4 \text{ kips})(5.06 \text{ ft}) \\ &= -319.4 \text{ ft-k} \end{aligned}$$

at point #3:

$$V_{UL} = (-98.8 \text{ kips}) + (31.2 \text{ klf} - 11.68 \text{ klf})(11 \text{ ft}) = -98.8 \text{ k} + 214.7 \text{ k} = 115.9 \text{ kips}$$

$$V_{UR} = (115.9 \text{ kips}) - (148.4 \text{ kips}) = -32.5 \text{ kips}$$

$$M_u = (31.2 \text{ klf} - 11.68 \text{ klf})(15.33 \text{ ft})^2 / 2 - (84.0 \text{ kips})(14 \text{ ft}) - (99.4 \text{ kips})(11 \text{ ft}) = 24.3 \text{ ft-k}$$

COMBINED FOUNDATIONS

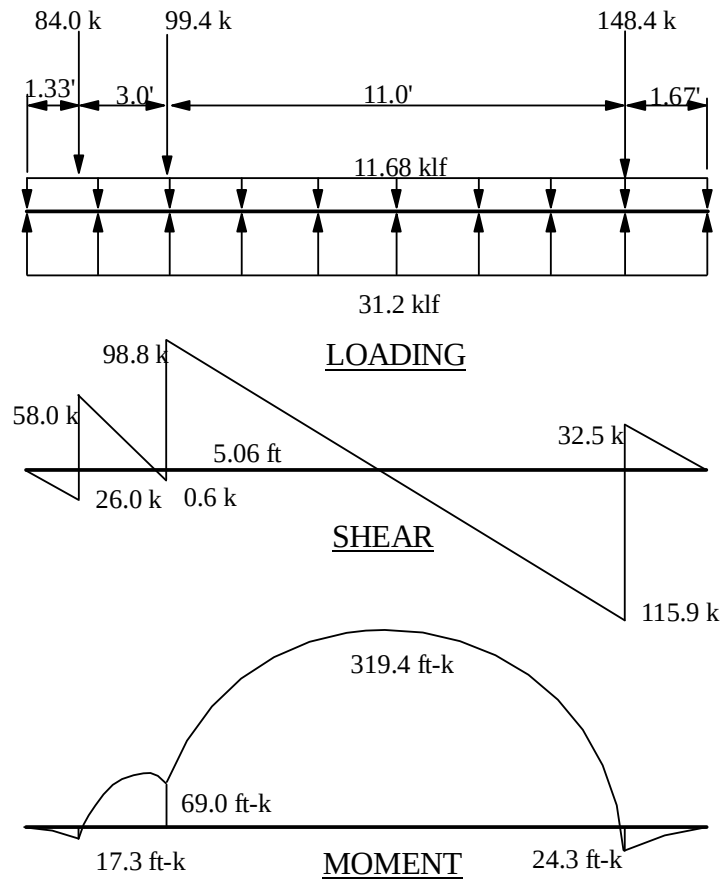
Sample Design of A Combined Foundation

check at right end:

$$V_U = (-32.5 \text{ kips}) + (31.2 \text{ klf} - 11.68 \text{ klf})(1.67 \text{ ft}) = 0.1 \text{ kips, close enough}$$

$$M_U = (24.3 \text{ ft-k}) - (31.2 \text{ klf} - 11.68 \text{ klf})(1.67 \text{ ft})^2 / 2 = -2.9 \text{ ft-k, close enough}$$

Case # 1, Operating:





COMBINED FOUNDATIONS

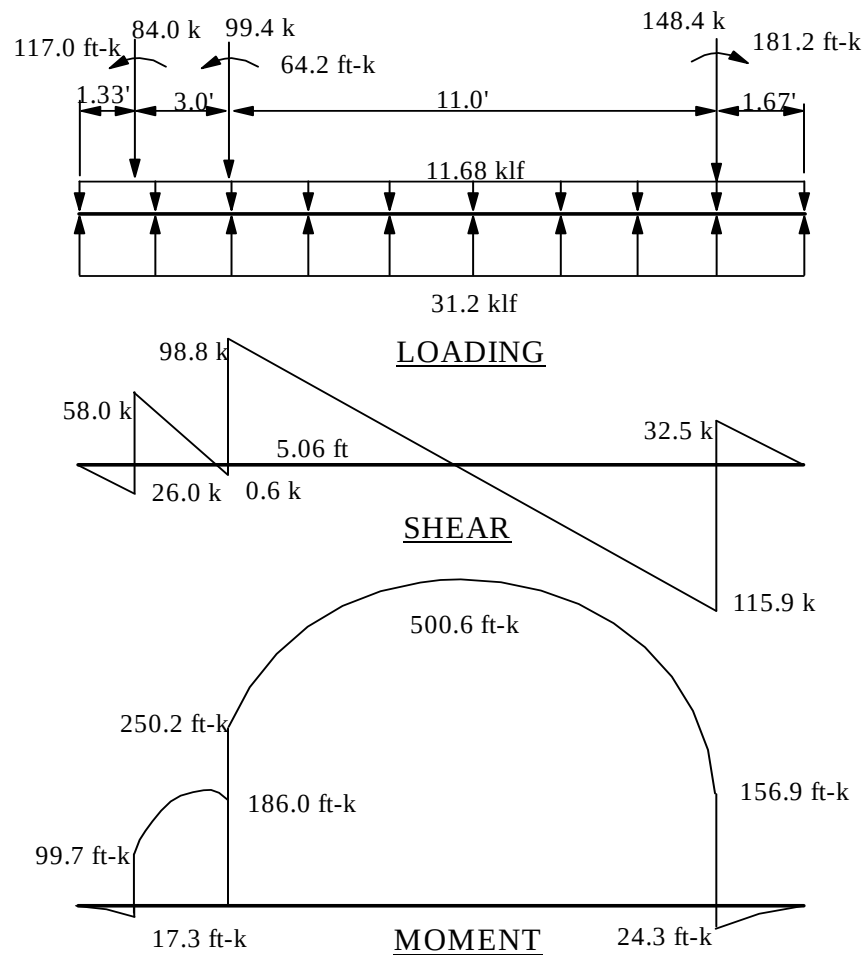
Sample Design of A Combined Foundation

Case #2, Operating + Friction

$$U = 1.4 D + 1.4 F_f$$

(Refer Structural Design Criteria, Table 6 – PIP STC 01015)

Calculations are omitted for brevity.

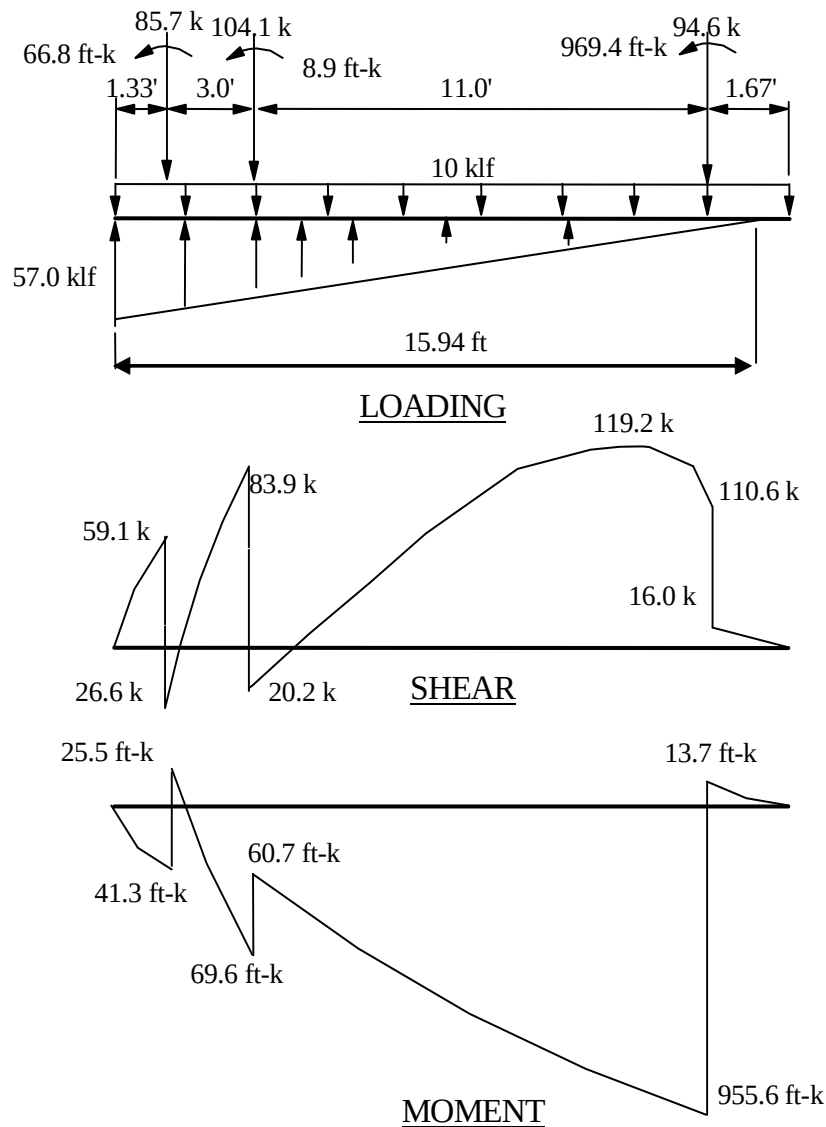


COMBINED FOUNDATIONS

Sample Design of A Combined Foundation

Case #3, Operating + Seismic (North to South)

U = 1.2D + E (Refer Structural Design Criteria, Table 6 – PIP STC 01015) Calculations are omitted for brevity.





COMBINED FOUNDATIONS

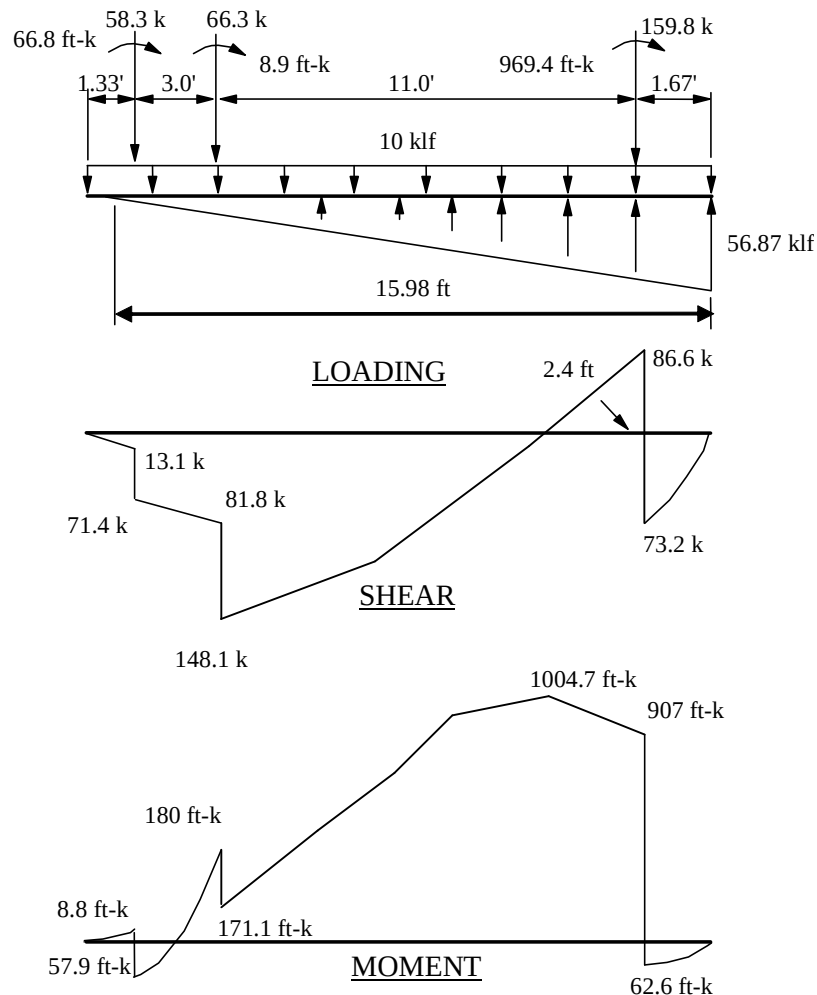
Sample Design of A Combined Foundation

Case #4. Operating + Seismic (South to North)

$U = 1.2D + E$

(Refer Structural Design Criteria, Table 6 – PIP STC 01015)

Calculations are omitted for brevity.



NOTE: In this example, engineering judgment was used to determine applicable load combinations to consider for design of the concrete and soil bearing. The Engineer must ensure all applicable load combinations are considered in design.

COMBINED FOUNDATIONS

Sample Design of A Combined Foundation

Concrete Design

Check Punching Shear

(Reference: ACI 318-05, and PCA Notes on ACI 318-05)

By inspection, the critical case is at the anchor end of the drum (at the corner of the mat).

Check stress for: Operating & Seismic, North to South

The following uses the ACI 318-05 strength load factor for seismic loads.

$$U = 1.2D + 1.0E$$

$$P_U = 1.2 (60 \text{ kips}) + 1.0 (-13.7 \text{ kips}) = 58.3 \text{ kips}$$

$$M_U = 1.0(39.1 \text{ kips})(2.17 \text{ ft}) + (597.9 \text{ ft-k}) = 682.7 \text{ ft-k}$$

$$d = (26 \text{ in mat}) - (3 \text{ in clear}) - (\text{say } 1.0 \text{ in bar})/2 = 22.5 \text{ in}$$

from PCA figure 16-13 Case D (Refer attachment 03), neglecting overhang of mat,

$$b_1 = c_1 + d/2 = (20 \text{ in}) + (22.5 \text{ in})/2 = 31.3 \text{ in}$$

$$b_2 = c_2 + d/2 = (30 \text{ in}) + (22.5 \text{ in})/2 = 41.3 \text{ in}$$

portion of pier moment resisted by mat moment (ACI equation 13-1),

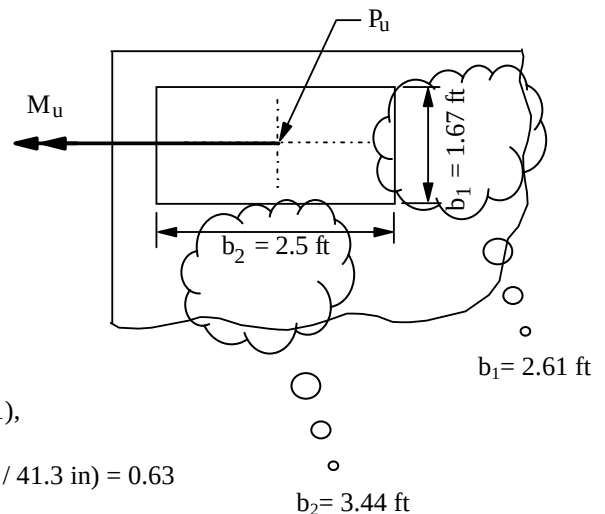
$$\gamma_f = 1 / [1 + (2/3)\text{sqrt}(b_1 / b_2)] = 1 / [1 + (2/3) \text{sqrt}(31.3 \text{ in} / 41.3 \text{ in})] = 0.63$$

portion of pier moment resisted by mat shear (ACI equation 11-39),

$$\gamma_v = 1 - \gamma_f = 1 - (0.63) = 0.37$$

section modulus of critical section (PCA figure 16-13 Case D) ,

$$\begin{aligned} J/C &= b_1^2 d (b_1 + 4b_2) + d^3 (b_1 + b_2) / 6b_1 \\ &= (31.3 \text{ in})^2 (22.5 \text{ in}) [(31.3 \text{ in}) + 4(41.3 \text{ in})] + (22.5 \text{ in})^3 [(31.3 \text{ in}) + (41.3 \text{ in})] / 6(31.3 \text{ in}) \\ &= 27468 \text{ in}^3 \end{aligned}$$





COMBINED FOUNDATIONS

Sample Design of A Combined Foundation

area of critical section, for corner column CASE D,

$$A_c = d (b_1 + b_2) = (22.5 \text{ in}) [(31.3 \text{ in}) + (41.3 \text{ in})] = 1634 \text{ in}^2$$

pier axial load, minus soil pressure, is resisted as mat shear, (in this case, soil pressure under pedestal is zero)

$$V_U = P_U (\text{pier}) - (SB_3)(\text{area}) = (58.3 \text{ kips}) - (0 \text{ ksf})(31.3 \text{ in} / 12)(41.3 \text{ in} / 12) = 58.3 \text{ kips}$$

maximum shear stress (ACI 318-05 Chapter R11.12.6.2),

$$\begin{aligned} V_{U1} &= V_U / A_c + \gamma_v (M_U) / (J/C) \\ &= (58.3 \text{ kips}) / (1634 \text{ in}^2) + (0.37)(682.7 \text{ ft-k} * 12) / (27468 \text{ in}^3) \\ &= 0.146 \text{ ksi} = 146 \text{ psi} \end{aligned}$$

determine allowable shear strength from ACI 318-05, Chapter 11.12.2.1,

case (a),

$$\beta_c = b_2 / b_1 = (41.3 \text{ in}) / (31.3 \text{ in}) = 1.32$$

$$V_c = \phi (2 + 4 / \beta_c) \sqrt{f'_c} b d = 0.75(2 + 4 / 1.32) \sqrt{4,000 \text{ psi}} b d = (239 \text{ psi}) b d \quad (\text{ACI eqn 11-33})$$

case (b),

$$\alpha_s = 20 \text{ (corner column)}$$

$$b_0 = b_1 + b_2 = (31.3 \text{ in}) + (41.3 \text{ in}) = 72.6 \text{ in}$$

$$V_c = \phi [\alpha_s d / b_0 + 2] \sqrt{f'_c} b d = 0.75[20 (22.5 \text{ in}) / (72.6 \text{ in}) + 2] \sqrt{4,000 \text{ psi}} b d = (389 \text{ psi}) b d \quad (\text{ACI eqn 11-34})$$

(for value of ϕ refer ACI 318-05 Chapter. 9.3.2.3)

case (c),

$$V_c = \phi (4) \sqrt{f'_c} b d = 0.75(4) \sqrt{4,000 \text{ psi}} b d = (189 \text{ psi}) b d \quad (\text{ACI eqn 11-35})$$

case (c) controls, $V_c = 189 \text{ psi}$. Check applied shear versus nominal,

$$V_{U1} = 146 \text{ psi} < V_c = 189 \text{ psi}, \quad \text{OK}$$



COMBINED FOUNDATIONS

Sample Design of A Combined Foundation

Check Mat Shear Capacity

maximum shear,

$$V_U = 148.1 \text{ kips} \quad (\text{operating plus seismic, south to north})$$

$$d = 22.5 \text{ in}$$

$$b = (14.5 \text{ ft})(12 \text{ in/ft}) = 174 \text{ in}$$

$$\phi V_N = \phi(V_C + V_S) \quad (\text{ACI eqn 11-1 \& 11-2})$$

V_C : Nominal shear strength provided by concrete

V_S : Nominal shear strength provided by shear reinforcement = 0

$$V_C = 2\sqrt{f'_c}bd \quad (\text{ACI eqn 11-3})$$

$$\phi V_N = [0.75(2)\sqrt{4,000 \text{ psi}}(174 \text{ in})(22.5 \text{ in}) + 0] / 1000 = 371.4 \text{ kips} > V_U \quad \text{OK}$$

Flexural Reinforcing (bottom steel)

maximum moment,

$$M_U = 955.6 \text{ ft-k} \quad (\text{operating plus seismic, north to south})$$

$$d = 22.5 \text{ in}$$

$$b = (14.5 \text{ ft})(12 \text{ in/ft}) = 174 \text{ in}$$

$$K_n = M_U / (\phi)(bd^2 / 12,000) = (955.6 \text{ ft-k}) / (0.9)[(174 \text{ in})(22.5 \text{ in})^2 / (12,000)] = 145$$

$$\phi = 0.9 \text{ (tension controlled section)}$$

from ACI Design Handbook SP-17, for $K_n = 145 \implies \rho = 0.0025$

check minimum reinforcing ratio (ACI 318-05 Chapter 7.12.2.1), $\rho_{\text{MIN}} = 0.0018 < 0.0025$, OK

$$A_s = \rho b d = (0.0025)(174 \text{ in})(22.5 \text{ in}) = 9.8 \text{ in}^2 \quad (0.68 \text{ in}^2 \text{ per foot}) \quad \text{USE \#8 BARS @ 10 IN}$$



COMBINED FOUNDATIONS

Sample Design of A Combined Foundation

Flexural Reinforcing (top steel)

maximum moment,

$$M_U = 1004.7 \text{ ft-k} \quad (\text{operating plus seismic, south to north})$$

$$d = 22.5 \text{ in}$$

$$b = (14.5 \text{ ft})(12 \text{ in/ft}) = 174 \text{ in}$$

$$K_n = M_U / (\phi)(bd^2 / 12000) = (1004.7 \text{ ft-k}) / (0.9)[(174 \text{ in})(22.5 \text{ in})^2 / (12,000)] = 152$$

from ACI design handbook SP-17, $\rho = 0.0027$

check minimum reinforcing ratio (ACI 318-05 Chapter 7.12.2.1), $\rho_{\text{MIN}} = 0.0018 < 0.0027$, OK,

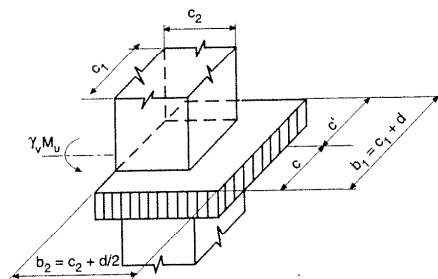
$$A_s = \rho b d = (0.0027)(174 \text{ in})(22.5 \text{ in}) = 10.57 \text{ in}^2 \quad (0.73 \text{ in}^2 \text{ per foot})$$

USE #8 BARS @ 10 IN

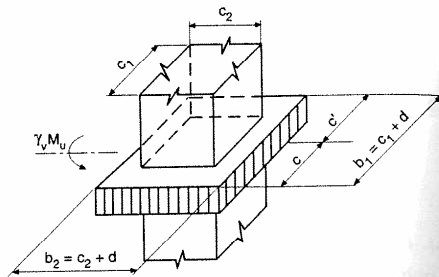
A similar analysis must be performed for the transverse (East -West) direction

FROM PCA NOTES – FIG 16.13

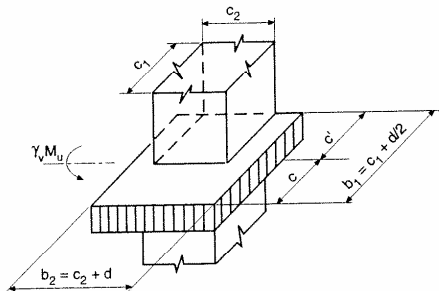
Case A: Edge Column (Bending parallel to edge)



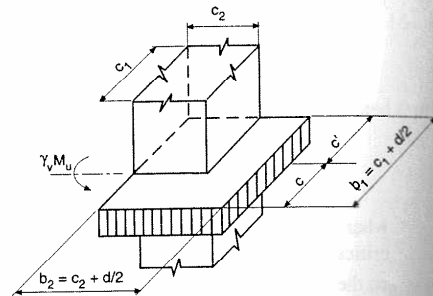
Case B: Interior Column



Case C: Edge Column (Bending perpendicular to edge)



Case D: Corner Column



Case	Area of critical section, A_c	Modulus of critical section		c	c'
		J/c	J/c'		
A	$(b_1 + 2b_2)d$	$\frac{b_1 d(b_1 + 6b_2) + d^3}{6}$	$\frac{b_1 d(b_1 + 6b_2) + d^3}{6}$	$\frac{b_1}{2}$	$\frac{b_1}{2}$
B	$2(b_1 + b_2)d$	$\frac{b_1 d(b_1 + 3b_2) + d^3}{3}$	$\frac{b_1 d(b_1 + 3b_2) + d^3}{3}$	$\frac{b_1}{2}$	$\frac{b_1}{2}$
C	$(2b_1 + b_2)d$	$\frac{2b_1^2 d(b_1 + 2b_2) + d^3(2b_1 + b_2)}{6b_1}$	$\frac{2b_1^2 d(b_1 + 2b_2) + d^3(2b_1 + b_2)}{6(b_1 + b_2)}$	$\frac{b_1^2}{2b_1 + b_2}$	$\frac{b_1(b_1 + b_2)}{2b_1 + b_2}$
D	$(b_1 + b_2)d$	$\frac{b_1^2 d(b_1 + 4b_2) + d^3(b_1 + b_2)}{6b_1}$	$\frac{b_1^2 d(b_1 + 4b_2) + d^3(b_1 + b_2)}{6(b_1 + 2b_2)}$	$\frac{b_1^2}{2(b_1 + b_2)}$	$\frac{b_1(b_1 + 2b_2)}{2(b_1 + b_2)}$

Fig. 16-13 Section Properties for Shear Stress Computations – Rectangular Columns